

ERA Tree Search vs. Best-of- N Sampling: A Small Local Reproduction

Reproduction Experiment Log — Stage 2

June 29, 2026

1 Goal

This is the second stage of reproducing the official [google-research/era](#) demo (for the Nature paper *An AI system to help scientists write expert-level empirical software*). Stage 1 confirmed that the ERA tree-search loop runs and improves a solution over iterations. Stage 2 tests a sharper claim from the paper: that **ERA is not merely best-of- N sampling** — its tree search is supposed to explore the space of programs *more efficiently* than drawing N independent samples and keeping the best.

2 Experimental Setup

Both methods run on the **same** task, the same local train/test split, and the same scoring function as `playground_s3e1.py` (the California-Housing-style regression demo). The metric is the **negative RMSE** ($-\text{RMSE}$); higher (closer to 0) is better. Both methods start from the identical initial solution (a plain linear regression) and are given an **equal budget of $N = 20$ LLM calls**.

- **ERA tree search (Method A).** Runs the existing FUTS / ERA search for 20 expansions. Each new candidate is generated conditioned on a *tree-selected parent*, so promising solutions are built upon.
- **Best-of- N independent sampling (Method B).** Generates 20 candidates, each conditioned on the *same* initial solution and score. It never conditions on previously generated candidates — every sample is drawn independently from the identical starter prompt, making it a fair independent-sampling baseline.

Because both methods make exactly N LLM calls, the x -axis (“number of candidate programs evaluated”) is directly comparable.

3 Results

Table 1 summarises the run, and Figure 1 shows the running-best score versus the number of candidates evaluated.

4 Interpretation

- **ERA wins under an equal budget.** With the same 20 LLM calls, ERA reached -0.5735 while best-of- N plateaued at -0.6149 , a gap of about 0.041 in RMSE. ERA also keeps improving late in the run, whereas best-of- N flattens early.

Quantity	Best score (negative RMSE, higher is better)
Initial baseline	-0.7339
ERA tree search (final best)	-0.5735
Best-of- N (final best)	-0.6149
Winner	ERA tree search

Table 1: Final results after $N = 20$ candidates per method (single run).

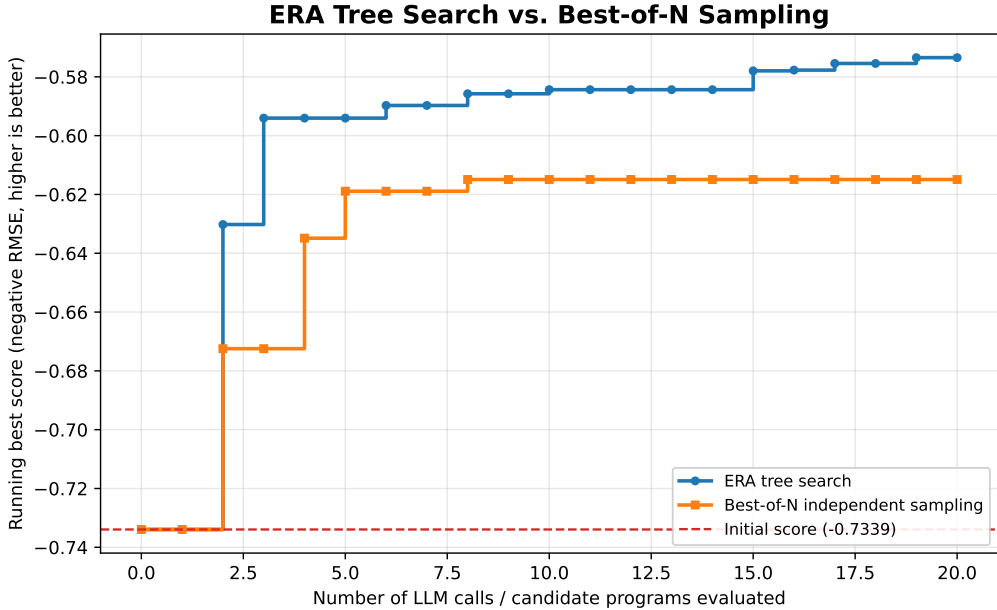


Figure 1: Running-best score (negative RMSE) versus number of candidate programs evaluated. Blue: ERA tree search. Orange: best-of- N independent sampling. Red dashed line: the shared initial score. Under an equal budget of 20 LLM calls, ERA reaches a clearly better solution.

- **Conditioning on good parents helps twice.** ERA generates each new candidate from the *best working solution so far*, so it both (a) climbs higher and (b) is far more *reliable*: in this run ERA produced only 3/20 invalid (non-executing) candidates, versus 13/20 for best-of- N . Best-of- N repeatedly tries to write a complex feature-engineered model from scratch off the weak linear baseline, and fails more often; ERA instead edits code that already runs.
- **Consistent with the paper’s claim.** This miniature experiment reproduces, qualitatively, the idea that ERA’s tree search is more sample-efficient than naive independent sampling — it is “not just best-of- N .”

5 Caveats and Limitations

- **Single stochastic run.** This is one run with $N = 20$. LLM sampling is random, so the exact gap will vary; a rigorous claim should average several seeds and/or use a larger N with confidence intervals.
- **Toy task.** The California-Housing demo has a small search space and plateaus quickly, so absolute gaps are modest.

- **Insecure sandbox.** Candidates are executed via the local `sandbox.py`, which is *not* a secure sandbox (it runs LLM-generated code directly). It is used here only for this trusted toy reproduction.

6 Next Steps

- Repeat the comparison over several random seeds and report mean $\pm$ standard deviation of the final best score for each method.
- Sweep the budget N (e.g. 10, 20, 40) to see how the ERA-vs-best-of- N gap scales with the number of LLM calls.
- Once the gap is robust on the toy task, move toward the fuller scientific tasks from the paper.